

CSA 1718 – Artificial Intelligence

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Prolog Program for Fruit and Its Color Using Backtracking

Aim

To write and implement a Prolog program to find fruits and their corresponding colors using backtracking.

Objective

- To understand backtracking in Prolog.
- To store fruit-color relationships.
- To obtain multiple solutions using repeated queries.

Theory

Backtracking is an important feature of Prolog. When multiple facts satisfy a query, Prolog gives one solution at a time. By pressing ;, the user can ask Prolog to search for the next solution.

Algorithm

1. Define different fruits and their colors.
2. Use the fruit/2 predicate.
3. Query fruit(Fruit, Color).
4. Prolog returns the first matching fruit and color.
5. Press ; to obtain the next solution.

Program

```
fruit(apple, red).
```

```
fruit(banana, yellow).
```

```
fruit(mango, yellow).
```

```
fruit(orange, orange).
```

```
fruit(grapes, green).
```

```
fruit(watermelon, green).
```

Query

fruit(Fruit, Color).

Output

Fruit = apple,

Color = red ;

Fruit = banana,

Color = yellow ;

Fruit = mango,

Color = yellow ;

Fruit = orange,

Color = orange ;

Fruit = grapes,

Color = green ;

Fruit = watermelon,

Color = green.

```
Program +
1 fruit(apple, red).
2 fruit(banana, yellow).
3 fruit(mango, yellow).
4 fruit(orange, orange).
5 fruit(grapes, green).
6 fruit(watermelon, green).
```



fruit(Fruit, Color).

Color = red,
Fruit = apple

Next 10 100 1,000 Stop

?- fruit(Fruit, color).

Result

Thus, the fruit and color information was successfully retrieved using Prolog backtracking.